

Uğur Emin Baynal *Electrical & Electronics/Computer Engineering Student*

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🎓 EDUCATION

Electrical and Electronics Engineering,
Izmir University of Economics

Oct 2021 – present | Izmir, Türkiye
Current GPA: 3.16/4

Computer Engineering, *Izmir University of Economics*

Sep 2022 – present | Izmir, Türkiye
Current GPA: 2.58/4

Math and Science, *Ankara Gazi Anatolian High School*

Sep 2017 – Jun 2021 | Ankara, Türkiye
93.04% graduation score.

💼 PROFESSIONAL EXPERIENCE

Software Development Specialist, *İzmir University of Economics*

Jan 2026 – Present | İzmir, Türkiye

R&D Engineering Intern, *EPS Technology*

Jun 2025 – Sep 2025 | Izmir, Türkiye

- Supported an R&D project focused on color science and image processing
- Contributed to experimental setups and data evaluation processes

Undergraduate Teaching Assistant, *Izmir University of Economics
Electrical and Electronics Engineering Department*

Oct 2024 – 2025 | Izmir, Turkey

- Assisted in lab courses involving OrCAD PSpice, circuit analysis, and embedded systems
- Helped students understand theoretical and practical components of engineering subjects

Engineering Tutor, *Pi-Course*

Aug 2024 – Present

- Delivered personalized instruction in electrical/electronics and software courses
- Prepared students for midterms and finals with tailored resources

Electronics Engineering Intern, *ASELSAN Presicion Optics*

Aug 2024 – Sep 2024 | Turkey

Admittance-Quality and Thermal/Night Vision Departments

Full-Stack Developer, *EvolveChat*

Jan 2024 – Feb 2024 | Remote

I was responsible for the development of the company's AI products.

🧠 SKILLS

MATLAB/Simulink | Python |
Multi-Agent LLM Workflow Design | OrCAD | KiCAD |
Java | ROS 2 | SQL | CST Studio | Communication |
Organisation | Creativity | Teaching Skills | Teamwork |
Problem-Solving | Project Management

🏠 ORGANIZATIONS

IEEE IEU Robotics and Automation Society, *Chairman*

Jun 2025 – Present

IUE Advanced and Innovative Technologies Group, *Member*

Sep 2025 – Present

IUE Signal Processing Group, *Member*

Jun 2025 – Present

EcoFusion'24 by IEEE IUE Student Branch,
Arduino Workshop Instructor

Oct 2024 | Izmir, Türkiye

📁 PROJECTS

Silkworm, Co-Founder, Head of R&D

Sep 2025 – present

Designed and built a 3D plastic recycling machine, including mechanical structure and motor/heating controls. Developed microcontroller-based automation for temperature, motor speed, and safety functions. Tested and optimized the system for stable and efficient recycling performance.

K.A.Y.R.A., Know About Your Route Always

Sep 2024

BLE-Based Localisation and Wayfinding: A Scalable Approach for Complex Indoor Environments (TÜBİTAK 2204-A Supported)

Teknofest 2025 - ECOWHEELS, The aim of the competition is to design the most energy efficient electric vehicle.

Mar 2025

My responsibility included the design of the Vehicle Control System, focusing on the firmware of the said system to enable user interaction and modulate the car functions accordingly.

ECO ROVER, Software Team Lead / Electronics Team Member

Oct 2024

- Coordinated a multidisciplinary software team for an autonomous rover project
- Designed PCBs and integrated sensors using KiCAD and Altium Designer

A.Q.U.A. F.A.R.M., Automated Quality and Utility-Assisted Farming and Resource Management

Oct 2024 – Jan 2025

Multi-Disciplinary Engineering Project with Multi-Agent LLM Workflow for automated, soilless farming systems.

Teknofest 2024 Rocket Competition, Medium Altitude Category

Jan 2024 – Jul 2024

I was a member of the avionics team of GARUDA Rocket and responsible for electronic circuit design and component selection.

CycLock, Bicycle Security System Concept

Feb 2022 – Jun 2022

Ses Kes, Medical Noise Cancellation Device for Neurodivergent People

Sep 2023

Electronics and Computer Engineer/Project Lead

📖 COURSES

CE 475 - Fundamentals and Applications of Machine Learning,
Izmir University of Economics Engineering Faculty

In this course, I expanded my knowledge of machine learning algorithms. The course project was on a given dataset and expected to predict the outcomes via a coding environment of our choice. My implementation was on MATLAB.

MCE 412 - Autonomous Robotics, *Izmir University of Economics*

In this course, I learned fundamental methods of design and analysis on autonomous robotics.

EEE 453 - Adaptive Filters, *İzmir University of Economics*

In this course, I have gained knowledge on modeling random processes that generate random signals, stationary processes, linear optimal (Wiener) filtering, linear adaptive filtering, steepest descent, LMS, and RLS learning algorithms, and Kalman filter theory.

EEE 461 - Embedded System Design, *İzmir University of Economics*

CE 490 - Digital Image Processing, *İzmir University of Economics*

📜 CERTIFICATES

InovaTim AI Course, Izmir University of Economics Mentor Programme, MATLAB Onramp, MATLAB Signal Generation and Resampling, MATLAB Spectral Analysis Techniques, Simulink Onramp

**Model United Nations, Delegate, President Chair,
Under-Secretary-General, Director-General, Secretary-General,
Organisational Advisor**

Jan 2018 – Jun 2023

Model United Nations (MUN) is a type of conference
which aims to teach students how the UN works.

LANGUAGES

Turkish (Native) | **English** (Fluent) | **Spanish** (Beginner) |
German (Beginner) | **Latin** (Beginner)